

SiliconMind



SiliconMind-V1

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AI Engine for IC Design

Generate · Test · Debug Verilog/RTL Locally



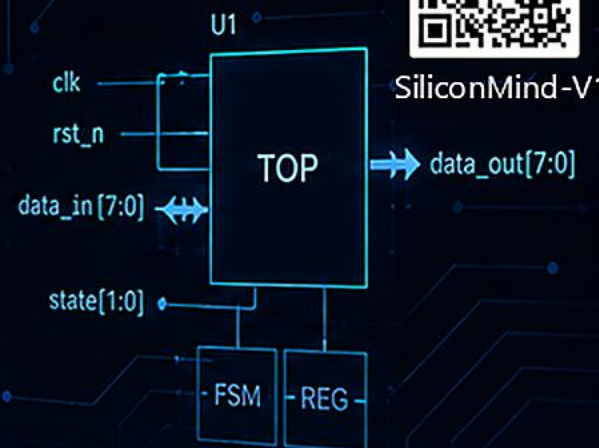
Academia Sinica & NTU

SiliconMind

Why IC Design Needs Private AI

Keep your design data and silicon secure

```
module top (  
    input clk,  
    input rst_n,  
    input [7:0] data_in,  
    output reg [7:0] data_out;  
  
    //FSM control  
    always @(posedge clk or negedge rst_n) begin  
        if (!rst_n)  
            state <= S_IDLE;  
        else  
            state <= next_state;  
        end  
    end  
endmodule
```



Sensitive IP

RTL code and
IP libraries are
confidential



Cloud Risk

Cloud LLMs may
expose source code
and inference logs



Correctness Matters

Syntax-correct
Verilog is not
enough

Generate · Test · Debug Verilog/RTL Locally

SiliconMind

SiliconMind-V1 Framework

From Spec to Verified Verilog / RTL



SiliconMind-V1

Multi-Agent
Training Data Pipeline



Produces reasoning-rich
verified training data

training data (used to train the model)

User Spec



SiliconMind
Inference Engine



Verified
Verilog / RTL

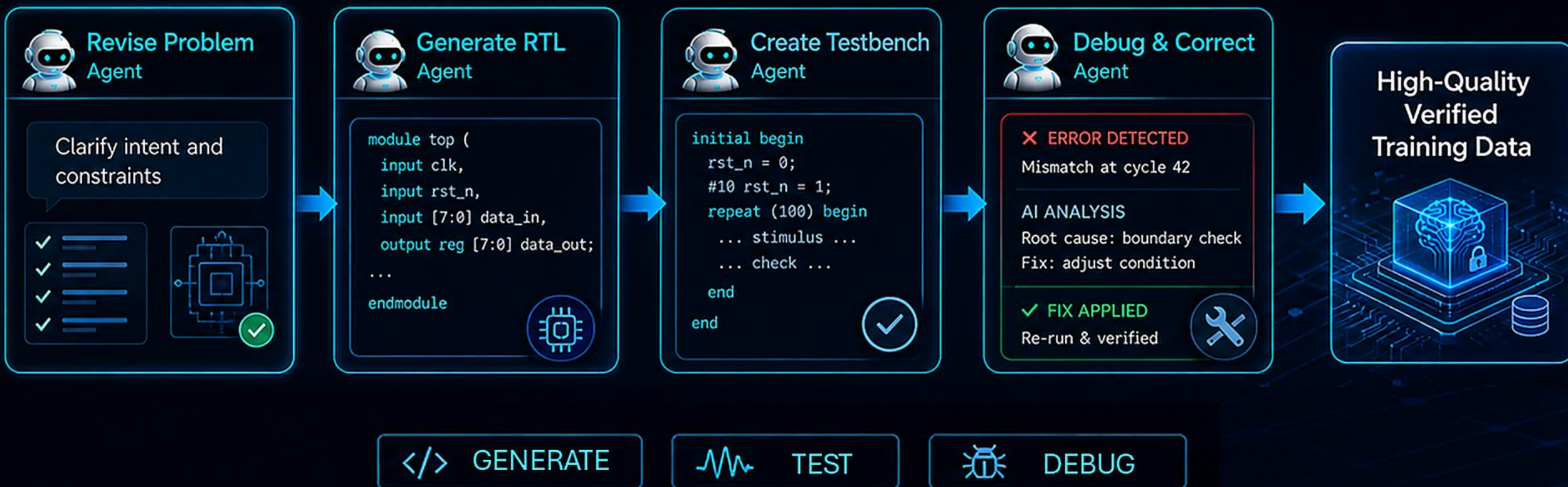


Generate · Test · Debug · Local · Private · Secure



Multi-Agent Distillation Pipeline

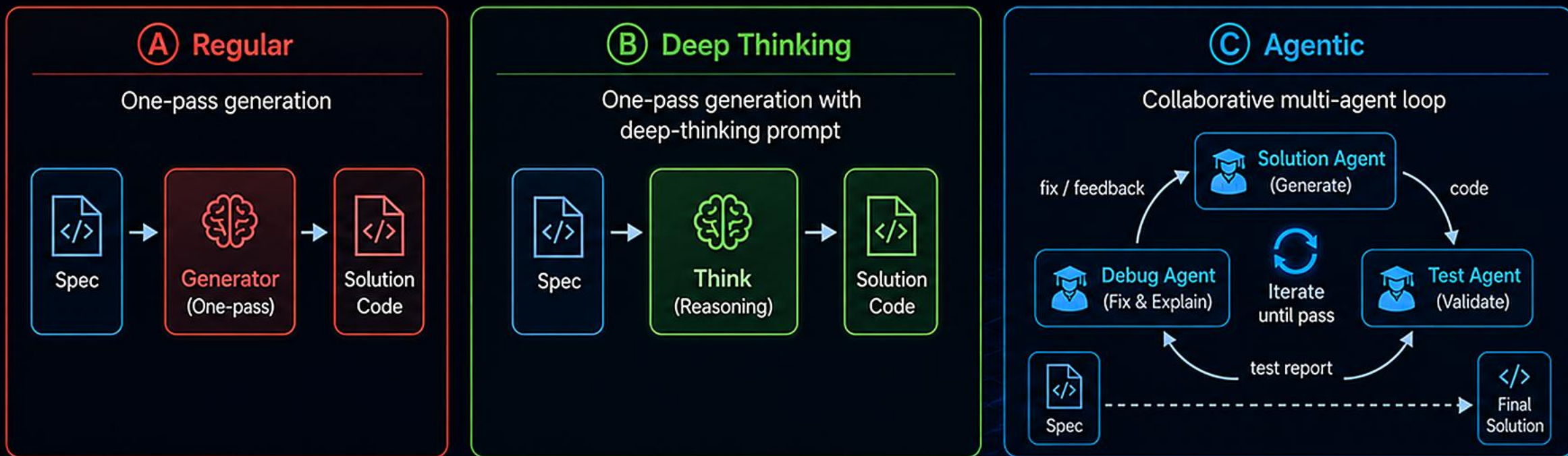
Teach the model to generate, test, and fix





From Code Generation to Debug-Reasoning

Generate, test, and debug with test-time scaling



Higher Accuracy
Fewer bugs, better results



Test-Time Scaling
More thinking, better code



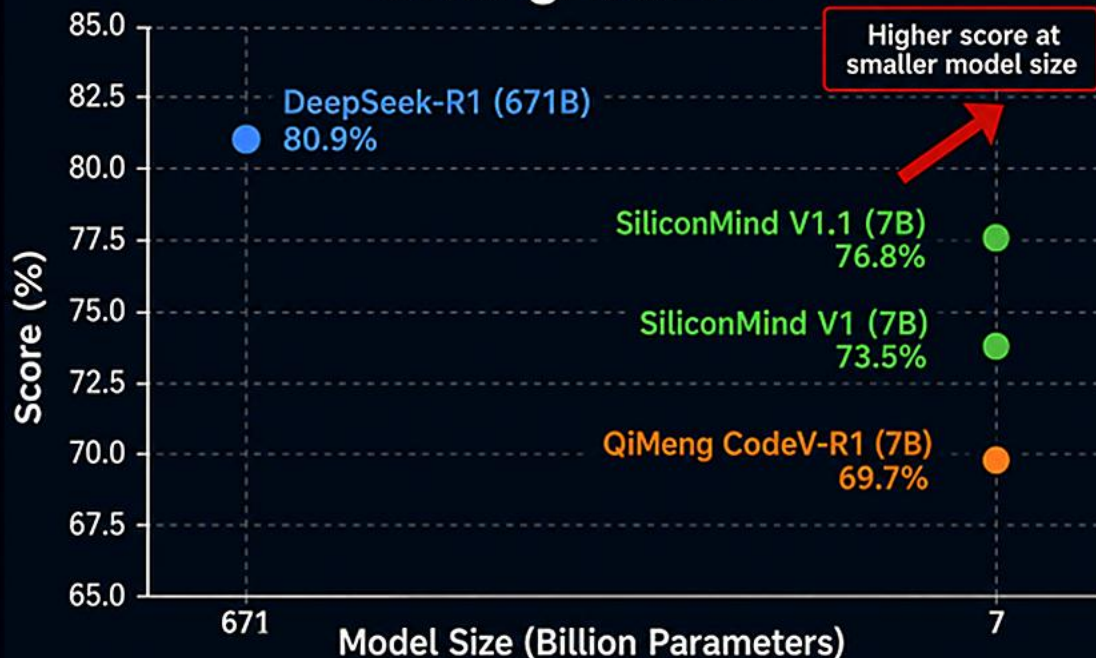
Robust & Reliable
Self-debug reduces errors



Faster to Deploy
Less human-in-the-loop



VerilogEval-v2



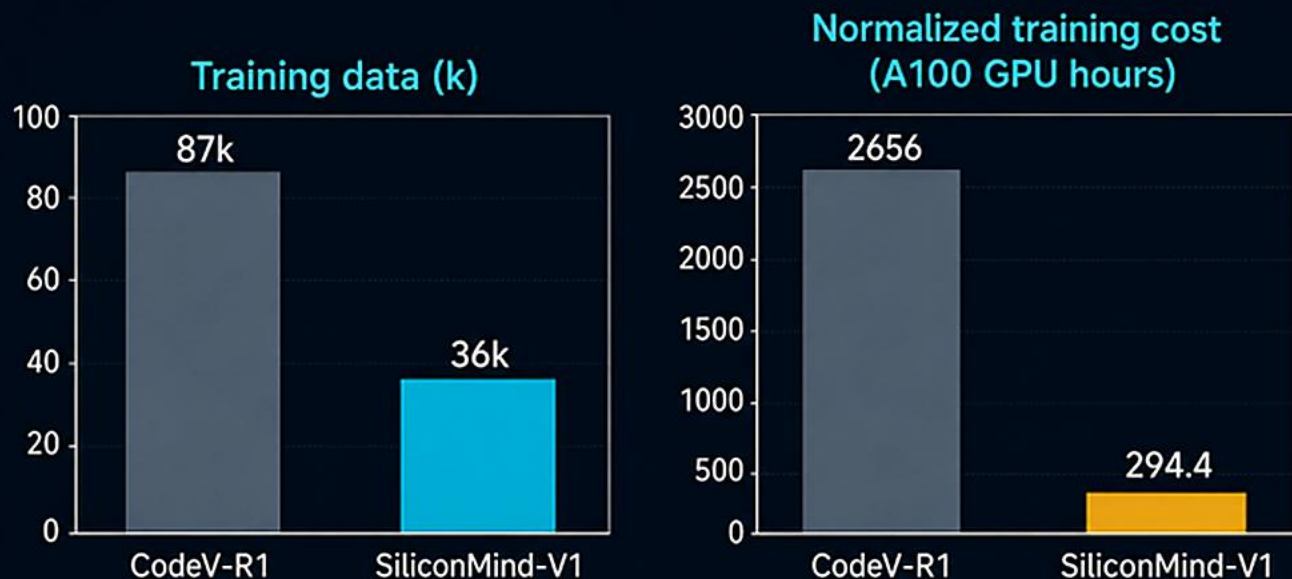
Smaller models are generally more parameter-efficient



Surpassing Previous SOTA

SiliconMind-V1 outperforms QiMeng-CodeV-R1, a model released by the Chinese Academy of Sciences, across multiple benchmarks, while narrowing the gap with DeepSeek-R1, which has over **95x** more parameters.

Lightweight training pipeline



High Training Efficiency

Compared with QiMeng-CodeV-R1, SiliconMind-V1 uses only **50%** of the training data and **1/9** of the compute while achieving comparable or better results.